THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 08

Full Syllabus (Chapters 1–14) | Class 10 Mathematics | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(c)** Incorrect — HCF ≤ LCM always	HCF divides LCM
2	**(c)** Incorrect — $\frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$ is irrational	Numerator/denominator must be integers
3	**(b)** −16	$\alpha\beta(\alpha+\beta) = -8 \times 2$
4	**(b)** 2	$4 + 2a - 8 = 0$
5	**(d)** $(\frac{3}{2}, 4)$	Ratios $= \frac{1}{2}$
6	**(c)** infinitely many solutions	Coincident lines
7	**(a)** real and equal	$D = 144 - 144 = 0$
8	**(a)** 9	$36 - 4k = 0$
9	**(c)** Incorrect — $d = 0$	$3 - 3 = 0$
10	**(a)** Student A is correct	Coefficient of n
11	**(a)** 2 cm	$\frac{DE}{6} = \frac{1}{3}$
12	**(d)** similar	Same shape, possibly different size
13	**(b)** Incorrect — $\frac{\sqrt{3}}{2} \neq 1$	$\sin 2A \neq 2\sin A$
14	**(c)** 1	Definition
15	**(b)** $\frac{77}{3}$ cm²	$\frac{1}{6} \times 154$
16	**(a)** 30°	6° per minute
17	**(b)** Incorrect — it is $\pi rl + 2\pi r^2$	No flat face is exposed
18	**(b)** 3	Most frequent
19	**(d)** 17	$\frac{90-22}{4}$
20	**(b)** Incorrect — they come closer as trials increase	Concept

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	$867 = 3 \times 17^2$, $255 = 3 \times 5 \times 17 \Rightarrow$ HCF $= 51$	1+1
22	$(x - \sqrt{3})(x - 1) \Rightarrow \sqrt{3}$ and 1	2
23	$(x-a)^2 = b^2 \Rightarrow x = a + b$ or $a - b$	1+1
24	$AB = 6$ cm, $BC = 8$ cm	1+1
25	$\frac{3+k}{2} = 4 \Rightarrow k = 5$	2
26	$\frac{1}{3}$	2
27	$30\sqrt{3} \approx 51.96$ m	2
28	$\angle PTQ = 70^\circ$; $TP = TQ \Rightarrow \angle TPQ = 55^\circ$	1+1

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Points $(0,5), (5,0)$; $(1,0), (0,-1)$. Lines intersect at $(3,2) \Rightarrow$ consistent, $x = 3$, $y = 2$	2+1
30	$a = 60, h = 20$; $\Sigma fu = -16 - 12 + 0 + 6 + 8 = -14 \Rightarrow 60 - \frac{280}{50} = 54.4$	1+1+1
31	$306, \ldots, 693$: $n = 44$; $S = 22(306 + 693) = 21978$	1+1+1
32	Lines touch at P, Q; $OP = OQ$, $\angle OPA = \angle OQA = 90^\circ$, OA common $\Rightarrow \triangle OPA \cong \triangle OQA$ (RHS) $\Rightarrow \angle OAP = \angle OAQ$	1+1+1
33	$18.84 - 15.57 = 3.27$ cm²	1+1+1
34	$\frac{1}{6}$; $\frac{5}{36}$; $\frac{11}{36}$	1+1+1
35	$h = x\sqrt{3} = \frac{x+40}{\sqrt{3}} \Rightarrow x = 20$ m (width); $h = 20\sqrt{3} \approx 34.64$ m	1+1+1

	OR: $d = 10\sqrt{3} \approx 17.32$ m; flagstaff $= d - 10 = 10(\sqrt{3} - 1) \approx 7.32$ m	1+1+1
36	$n = 50$; cf 4, 14, 30 ⇒ $30 + \frac{25-14}{16} \times 5 \approx 33.44$	1+1+1
	OR: $5 + \frac{15-9}{30-9-9} \times 2 = 6$ — the student is correct	1+1+1

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	$\frac{5k-5}{k+1} = 0 \Rightarrow k = 1$ ⇒ 1 : 1 (2). Point $= \left(\frac{-4+1}{2}, 0\right) = \left(-\frac{3}{2}, 0\right)$ (2); it is the mid-point ✓ (1)	2+2+1
	OR: $P = \left(\frac{24+6}{5}, \frac{14-9}{5}\right) = (6, 1)$ (2½). New A′(6, −3), B′(12, 2): $P' = \left(\frac{24+18}{5}, \frac{4-9}{5}\right) = \left(\frac{42}{5}, -1\right)$ (2½)	2½+2½
38	LHS $= \frac{\cos^2 A}{\sin A} \cdot \frac{\sin^2 A}{\cos A} = \sin A \cos A$; RHS $= \frac{1}{\frac{\sin^2 A + \cos^2 A}{\sin A \cos A}} = \sin A \cos A$ ✓	2+2+1
	OR: Write in $\sin, \cos$: $\frac{\sin^2\theta}{\cos\theta(\sin\theta - \cos\theta)} - \frac{\cos^2\theta}{\sin\theta(\sin\theta - \cos\theta)} = \frac{\sin^3\theta - \cos^3\theta}{\sin\theta\cos\theta(\sin\theta - \cos\theta)} = \frac{1 + \sin\theta\cos\theta}{\sin\theta\cos\theta} = 1 + \sec\theta\csc\theta$	2+2+1
39	Cylinder length 72: $2\pi rh \approx 8146.29$; sphere $4\pi r^2 \approx 4073.14$; total ≈ 12219.43 cm²; cost $\approx$ ₹855.36	2+2+1
	OR: (i) $\pi r(2h + l) = \frac{22}{7} \times 52.5 \times 59 = 9735$ m² (ii) ₹116820	3+2
40	(i) $\frac{1080}{40} = 27$ (ii) cf 4, 12, 22 ⇒ $20 + \frac{20-12}{10} \times 10 = 28$ (iii) $3(28) - 2(27) = 30$	2+2+1
	OR: (i) Modal 60–80: $60 + \frac{12-10}{24-10-6} \times 20 = 65$ (ii) $n = 50$, cf 6, 14, 24 ⇒ $60 + \frac{25-24}{12} \times 20 = 61.67$ (iii) No — he likely took the wrong modal class	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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